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increases from 5 cm^2 to 10 cm^2; 5 cm^2 in a given time. Leaf B increases from 50 cm^2 to 55 cm^2 ; 5 cm^2 in a given time. Hence, both leaves A and B increase their area by 5 cm^2 in a given time. This is absolute growth. Relative growth is faster in leaf A because of initial small size. It decreases with time (Figure 15.7).

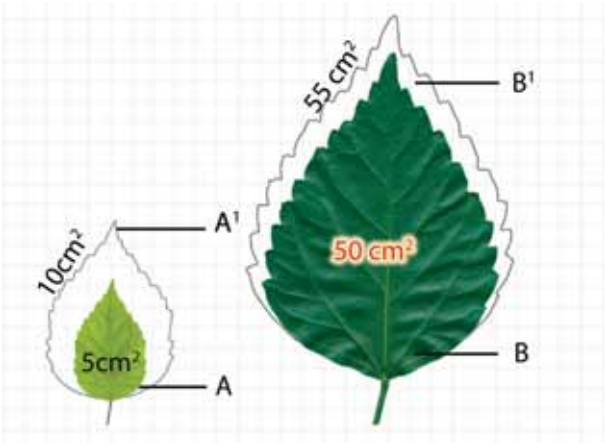

Figure 15.7: Diagrammatic comparision of absolute and relative growth rates

3.Conditions of growth

Plant growth is influenced by a variety of external and internal factors. A brief account of these factors is given below:

I. External Factors

a. Water

Water is essential for cell enlargement as well as growth in the size of the cell. Turgidity of cells helps in growth extension. Water provides the medium for enzymatic activities needed for growth.

b. Nutrition

Nutrition plays an important role in the formation of protoplasm. Macro and micro elements are very important as sources of energy. For example, carbon and oxygen in carbon-di-oxide and hydrogen in water are assimilated in photosynthesis.

c. Temperature

Temperature plays a significant role in the growth of the plant. Proper growth of a plant occurs at a about 28° C to 30° C temperature and above 45° C will damage the protoplasm and hinders the growth.

d. Oxygen

Oxygen has a vital role in the growth of the plant. It helps in releasing metabolic energy essential for growth activities. It is necessary for respiration.

e. Light

Light has its own contribution in the growth of the plant. Light is important for growth and photosynthesis. Light stimulates healthy growth. Absence of light may lead to yellowish in colour. This is called **etiolation**.

II. Internal Factors

a. Genes are intracellular factors for growth.

b. Phytohormones are intracellular factors for growth. Example: auxin, gibberellin, cytokinin.

c. C/N ratio.

The ratio of carbohydrates and nitrogenous compounds regulate the specific pattern of growth in plants. For example, if a plant contains more nitrogenous compounds as compared to carbohydrates it produces more protoplasm less mechanical tissues and vigorous vegetative growth. On the other hand, less nitrogenous compounds and more carbohydrates favour the synthesis of more wall material, less protoplasm, and more mechanical tissues.